MAT 3007 – Optimization

Complexity and Duality

Lecture 11

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Recap

Simplex Tableau:

- First identify an initial BFS with basis B and setup the canonical form:

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

- This is easy when $\boldsymbol{A}_B = \boldsymbol{I}$ and $\boldsymbol{c}_B = \boldsymbol{0}$ (which happens when we bring an LP with $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$ and $\boldsymbol{b} \geq \boldsymbol{0}$ into standard form).
- Identify the pivot column (negative reduced costs) and the pivot row (MRT) using Bland's rule.
- Perform elementary row manipulations to change the basis.
- This can also be applied in the two-phase simplex method.

Three Steps

Step 1: By looking at reduced cost (upper-left tableau), choose an index j to enter to basis. j-th column is the pivot column.

Step 2 By MRT (ratio between lower-left tableau and j-th column), choose a basic index $B(\ell)$ to leave. ℓ-th row is the pivot row

Step 3: Perform elemantary row operations to make j-th column $\begin{pmatrix} 0 \\ \hline e_\ell \end{pmatrix}$. Then in the new basis $\bar{B}$, we have $\bar{B}(\ell) = j$.

The Two-Phase Method & Simplex Tableau

Two-Phase Method in Simplex Tableau

Original problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

Auxiliary problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},\boldsymbol{y}} \quad & \mathbb{1}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0} \end{aligned}$$

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Phase I: Solve the auxilary problem with simplex tableau

- $(\boldsymbol{x} = \boldsymbol{0}; \boldsymbol{y} = \boldsymbol{b})$ is an initial BFS for the auxilairy problem with $\boldsymbol{y}$ being basic variables. But it's not in cononical form.

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- $(\boldsymbol{x} = \boldsymbol{0}; \boldsymbol{y} = \boldsymbol{b})$ is an initial BFS for the auxilairy problem with $\boldsymbol{y}$ being basic variables. But it's not in cononical form.
- Compute upper-left tableau (reduced costs):
  - For basic variables, the reduced costs are 0.
  - For nonbasic variables, $\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j = -\mathbb{1}^\top \boldsymbol{A}_j$, so the j-th reduced cost is the negative of the sum of that column.

use c in the auxiliary problem.

- Compute upper-right tableau (negative objective): $-\mathbb{1}^\top \boldsymbol{b}$

Two-Phase Method in Simplex Tableau

Assume that Phase I of the simplex method has been completed, yielding an optimal solution $(\boldsymbol{x}^*, \mathbf{0})$ with an optimal value of 0.

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Phase II: Solve the original problem from $\boldsymbol{x}^*$

1. Drop all columns associated with the auxiliary variables.
2. Change the upper-left to the reduced costs of the original problem:

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j$$

use c in the original problem.

   and the upper-right to the current (negative) objective value.
3. Continue the simplex tableau to solve the original problem.

Two-Phase Method in Simplex Tableau

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   and the upper-right to the current (negative) objective value.
3. Continue the simplex tableau to solve the original problem.

If the output $\boldsymbol{y}^* = \mathbf{0}$ includes basic variables, substitute the basic indices in $\boldsymbol{y}^*$ with some indices of the original variables such that they can form a full basis. (How to do that with Tableau?)

Example

$$
\begin{array}{lrrrrcr}
\text{minimize} & x_1 & +x_2 & +x_3 & & & \\
\text{subject to} & x_1 & +2x_2 & +3x_3 & & = & 3 \\
 & & -4x_2 & -9x_3 & & = & -5 \\
 & & & +3x_3 & +x_4 & = & 1 \\
 & x_1, & & x_2, & x_3, \quad x_4 & \geq & 0
\end{array}
$$

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 & & & +3x_3 & +x_4 & = & 1 \\
 & x_1, & x_2, & x_3, & x_4 & \geq & 0
\end{array}
$$

First, make $\boldsymbol{b}$ positive and construct the auxiliary problem:

$$
\begin{array}{lrrrrrrrcl}
\text{minimize} & & & & & x_5 & +x_6 & +x_7 & & \\
\text{subject to} & x_1 & +2x_2 & +3x_3 & & +x_5 & & & = & 3 \\
 & & 4x_2 & +9x_3 & & & +x_6 & & = & 5 \\
 & & & +3x_3 & +x_4 & & & +x_7 & = & 1 \\
 & x_1, & x_2, & x_3, & x_4, & x_5, & x_6, & x_7 & \geq & 0
\end{array}
$$

Example Continued

Construct the initial tableau for the auxiliary problem

B	-1	-6	-15	-1	0	0	0	-9
5	1	2	3	0	1	0	0	3
6	0	4	9	0	0	1	0	5
7	0	0	3	1	0	0	1	1

Example Continued

Construct the initial tableau for the auxiliary problem

B	-1	-6	-15	-1	0	0	0	-9
5	1	2	3	0	1	0	0	3
6	0	4	9	0	0	1	0	5
7	0	0	3	1	0	0	1	1

Carry out the simplex method: Iteration 1

B	0	-4	-12	-1	1	0	0	-6
1	1	2	3	0	1	0	0	3
6	0	4	9	0	0	1	0	5
7	0	0	3	1	0	0	1	1

MTR
3/2
5/4

Example Continued

Iteration 2:

B	0	0	-3	-1	1	1	0	-1
1	1	0	-3/2	0	1	-1/2	0	1/2
2	0	1	9/4	0	0	1/4	0	5/4
7	0	0	3	1	0	0	1	1

$\frac{5}{4} / \frac{9}{4} = \frac{5}{4} \cdot \frac{1}{3}$

Example Continued

Iteration 2:

B	0	0	-3	-1	1	1	0	-1
1	1	0	-3/2	0	1	-1/2	0	1/2
2	0	1	9/4	0	0	1/4	0	5/4
7	0	0	3	1	0	0	1	1

Iteration 3:

B	0	0	0	0	1	1	1	0
1	1	0	0	1/2	1	-1/2	1/2	1
2	0	1	0	-3/4	0	1/4	-3/4	1/2
3	0	0	1	1/3	0	0	1/3	1/3

This is optimal for the auxiliary problem. $\boldsymbol{x} = (1; \frac{1}{2}; \frac{1}{3}; 0)$ is a BFS for the original problem ($B = \{1, 2, 3\}$).

Example Continued

B	0	0	0	0	1	1	1	0
1	1	0	0	1/2	1	-1/2	1/2	1
2	0	1	0	-3/4	0	1/4	-3/4	1/2
3	0	0	1	1/3	0	0	1/3	1/3

Drop all the columns for auxiliary variables. We recompute the reduced cost for the original problem:

$$\bar{\boldsymbol{c}}^\top = \boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A} = (0, 0, 0, -\tfrac{1}{12}).$$

use c in the original problem.

$=0$ for basic variable

and the objective value: $\frac{11}{6}$.

Now the simplex tableau becomes:

B	0	0	0	-1/12	-11/6
1	1	0	0	1/2	1
2	0	1	0	-3/4	1/2
3	0	0	1	1/3	1/3

$A_B^{-1}A_j$ is j-th column in the tableau, so we don't need to compute.

$\bar{c}_4 = 0 - 1^\top \begin{pmatrix} \frac{1}{2} \\ -3/4 \\ 1/3 \end{pmatrix} = -\frac{1}{12}$

Example Continued

Then we continue from the new simplex tableau:

B	0	0	0	-1/12	-11/6
1	1	0	0	1/2	1
2	0	1	0	-3/4	1/2
3	0	0	1	1/3	1/3

Example Continued

Then we continue from the new simplex tableau:

B	0	0	0	-1/12	-11/6
1	1	0	0	1/2	1
2	0	1	0	-3/4	1/2
3	0	0	1	1/3	1/3

The next pivot:

B	0	0	1/4	0	-7/4
1	1	0	-3/2	0	1/2
2	0	1	9/4	0	5/4
4	0	0	3	1	1

$\Rightarrow x_1 = \frac{1}{2}$

$\Rightarrow x_2 = 5/4$

$\Rightarrow x_4 = 1$

This is optimal and we can stop. The optimal solution is $\boldsymbol{x} = (\frac{1}{2}; \frac{5}{4}; 0; 1)$. The optimal value is $\frac{7}{4}$.

Simplex Method Summary

We have completed our discussions of the simplex method:

- The idea of the simplex method (search among the BFS and neighbors) and its justifications.
- The algebraic procedures.
- The simplex tableau.
- Other issues about the simplex method: initialization, degeneracy, etc.

Complexity Theory

How Efficient is the Simplex Method?

We have learned that the simplex method terminates within a finite number of iterations.

$\rightsquigarrow$ But finite could still be large ...

- We know that the number of iterations should not exceed the number of BFS, which is bounded by $C(n, m)$. But this is a large number!

To answer this question, we discuss some basics about complexity theory.

Complexity

We count the number of arithmetic operations (addition: $+$, subtraction: $-$, multiplication: $\cdot$, division: $\div$, comparisons, etc.) that an algorithm needs to solve a problem. This is called the complexity of the algorithm.

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When we analyze the complexity of an algorithm, we consider the worst-case instance for that algorithm:

- Worst-case instance: a specific problem input that causes an algorithm to perform the maximum number of operations.
- For a specific instance, an algorithm may finish much faster than its complexity suggests.

Big-O Notation

We write $g(n) \sim O(f(n))$ if and only if there exists nonnegative constants c_1 and c_2 such that $|g(n)| \leq c_1 f(n)$ for all $n \geq c_2$.

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Examples: Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, $\boldsymbol{B} \in \mathbb{R}^{n \times p}$

- $\boldsymbol{x} + \boldsymbol{y}$: $O(n)$
- $\boldsymbol{x}^\top \boldsymbol{y}$: $O(n)$
- $\boldsymbol{A}\boldsymbol{x}$: $O(mn)$
- $\boldsymbol{A}\boldsymbol{B}$: $O(mnp)$

① $x+y = \begin{pmatrix} x_1+y_1 \\ \vdots \\ x_n+y_n \end{pmatrix}$

→ n additions.

number of operations

$g(n) = n \sim O(n)$

② $x^T y = \sum_{i=1}^{n} x_i y_i$

→ n multiplications

$n-1$ additions.

→ number of operations

$g(n) = 2n-1 \sim O(n)$

③ $Ax = \begin{pmatrix} a_1^T x \\ a_2^T x \\ \vdots \\ a_m^T x \end{pmatrix}$

→ number of operations

$g(m,n) = m(2n-1)$

$\sim O(mn)$

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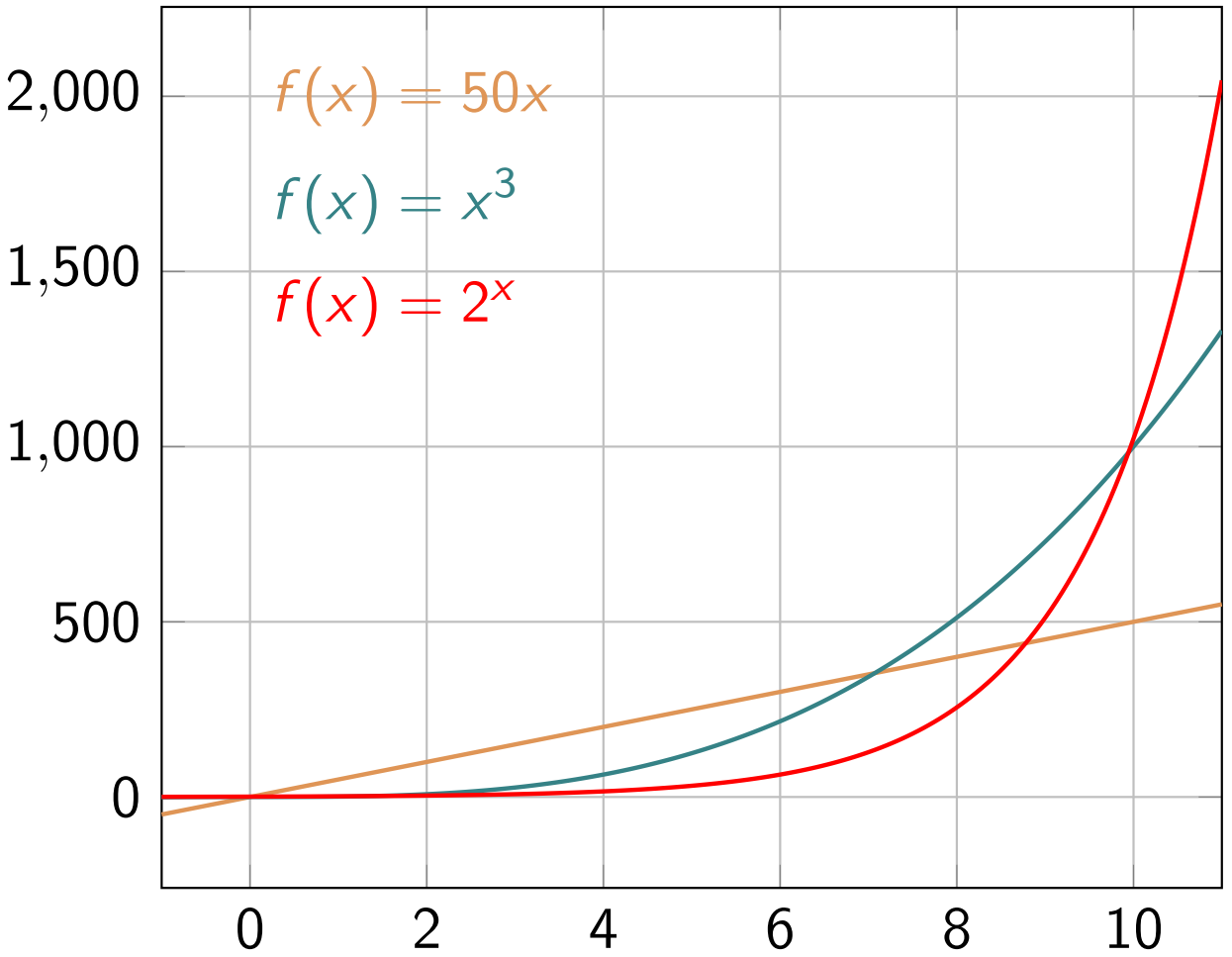
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The main watershed in complexity is whether an algorithm has a polynomial complexity or not:

- We call an algorithm a polynomial-time algorithm if the number of arithmetic operations required to solve any instance of the problem is bounded by a polynomial of the input size.
- A polynomial-time algorithm is usually deemed to be a practical algorithm.

Polynomial vs Non-Polynomial



- Polynomial vs Exponential Function.

Polynomial-Time Algorithms

Here are some known polynomial algorithms:

- Gaussian elimination for matrix inversion: $O(n^3)$.
- Fast method for matrix inversion: $\sim O(n^{2.373})$.
- Naive method for sorting n numbers: $O(n^2)$.
- Merge sort for sorting n numbers: $O(n \log n)$.

Here are some non-polynomial algorithms:

- Enumeration method for traveling salesman problem: $O(n!)$.
- Dynamic programming for TSP: $O(2^n n^2)$.

Complexity of a Problem

Definition: P

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There exists a class of problems that researchers haven't found any polynomial-time algorithm (e.g., the traveling salesman problem):

- This class is called NP-hard problems.
- In fact, if one can show that one NP-hard problem has a polynomial-time algorithm, then a large subclass of them will have polynomial-time algorithms.

Complexity of the Simplex Method

We consider a standard LP with n variables and m constraints.

What is the complexity of the simplex method?

- We can configure the simplex method by selecting different rules for the incoming/leaving basis indices. Each will result in a different algorithm and may have different complexity.

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For all the known simplex variants, problem instances have been found such that the algorithm will run with exponential iterations in terms of n and m to solve that problem (Klee and Minty Example).

- Very bad news from a theoretical point of view.

Complexity of Simplex Method

However ...

- This is only the worst-case performance.
- The practical performance of the simplex method is quite good. Typically, it needs a small multiple of m iterations.
- It is still one of the most widely-used algorithms for LPs.
- It is known to be polynomial-time on average (via "smoothed analysis").

Complexity of Linear Problems

Although the simplex method may not be a polynomial-time alg., this does not mean that LPs are not polynomial-time solvable.

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Whether LP is in P was a major problem in the 20th century. It was finally solved by Soviet Union mathematician Khachiyan in 1979, who showed a first polynomial-time algorithm for LPs. His method is called the ellipsoid method.

- Although the ellipsoid method is a polynomial-time algorithm (in a weak sense/in the Turing-machine model), it is extremely slow in practice. Therefore, it is more of a theoretical contribution than a practical one.

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Can we do better both theoretically and practically? Yes. The Interior Point Method. We will have a brief introduction later.

Duality Theory for LPs

Duality Theory: Motivation

We consider the standard LP (m constraints, n variables):

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

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if $b_i > a_i^\top x$, we let $y_i \to \infty$
if $b_i < a_i^\top x$, we let $y_i \to -\infty$.

We can write it as:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} + \max_{\boldsymbol{y} \in \mathbb{R}^m} \boldsymbol{y}^\top (\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

$= \sum_i y_i (b_i - a_i^\top x)$

Why?

- If $\boldsymbol{A}\boldsymbol{x} \neq \boldsymbol{b}$, then $\boldsymbol{y}^\top (\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}) \to \infty$ by choosing y, which can not be optimal. Hence, it implicitly enforces the constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. If $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, then $\boldsymbol{y}^\top (\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}) = 0$, so they attain the same objective value.

Continue . . .

$$\min_{\boldsymbol{x}\geq \mathbf{0}} \quad \max_{\boldsymbol{y}} \quad \boldsymbol{c}^\top \boldsymbol{x} + \boldsymbol{y}^\top (\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x})$$

$\rightsquigarrow$ Now we assume that we can exchange max and min (we will justify it later). Then the problem becomes:

$$\max_{\boldsymbol{y}} \quad \boldsymbol{b}^\top \boldsymbol{y} + \min_{\boldsymbol{x}\geq \mathbf{0}} \ \boldsymbol{x}^\top (\boldsymbol{c} - \boldsymbol{A}^\top \boldsymbol{y})$$

Continue . . .

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$$\max_{\boldsymbol{y}} \quad \boldsymbol{b}^\top \boldsymbol{y} + \min_{\boldsymbol{x}\geq \mathbf{0}} \; \boldsymbol{x}^\top (\boldsymbol{c} - \boldsymbol{A}^\top \boldsymbol{y})$$

We claim that this is equivalent to:

$$\begin{aligned} \text{maximize}_{\boldsymbol{y}} \quad & \boldsymbol{b}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}. \end{aligned}$$

Why?

$$\min_{\boldsymbol{x}\geq \mathbf{0}} \; \boldsymbol{x}^\top (\boldsymbol{c} - \boldsymbol{A}^\top \boldsymbol{y}) = \begin{cases} 0 & \text{if } \boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}, \\ -\infty & \text{if } \boldsymbol{A}^\top \boldsymbol{y} \not\leq \boldsymbol{c}. \end{cases}$$

Two Linear Programs . . .

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

and

$$\begin{aligned} \text{maximize}_{\boldsymbol{y}} \quad & \boldsymbol{b}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c} \end{aligned}$$

We call them dual to each other. If we call one the primal problem, then the other one is called the dual problem.

- We call $\boldsymbol{y}$ the dual variables (to the first LP).

Two Linear Programs . . .

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$\rightsquigarrow$ By our derivation, they should have the same optimal value (provided the exchange of min and max is valid).

Dual Problem

We have derived our first pair of dual problems.

- Duality theory is very important for linear optimization (and for more general optimization problems as well).
- The dual problem carries much useful information for the primal problem ($\rightsquigarrow$ later).
- The dual problem can also be used to solve the primal problem.

Different Duality Forms

We have derived the dual problem of the standard form. What if we want to find the dual problem of:

$$\begin{aligned} \text{minimize} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b}. \end{aligned} \tag{1}$$

One can first transform (1) to the standard form and then derive the dual. But we can directly apply our previous arguments.

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One can first transform (1) to the standard form and then derive the dual. But we can directly apply our previous arguments.

The primal problem can be written as:

$$\min_{\boldsymbol{x}} \quad \boldsymbol{c}^\top \boldsymbol{x} + \max_{\boldsymbol{y} \geq \boldsymbol{0}} \; \boldsymbol{y}^\top (\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x})$$

if $b_i > a_i^\top x$, let $y_i \to \infty$.

Why?

$$\max_{\boldsymbol{y} \geq \boldsymbol{0}} \; \boldsymbol{y}^\top (\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}) = \begin{cases} 0 & \text{if } \boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b}, \\ \infty & \text{if } \boldsymbol{A}\boldsymbol{x} \not\geq \boldsymbol{b}. \end{cases}$$

Assume we can exchange the order of max and min, then:

$$\max_{\boldsymbol{y}\geq \mathbf{0}} \quad \boldsymbol{b}^\top \boldsymbol{y} + \min_{\boldsymbol{x}} \ \boldsymbol{x}^\top (\boldsymbol{c} - \boldsymbol{A}^\top \boldsymbol{y}).$$

This is further equivalent to:

$$\begin{aligned} \text{maximize}_{\boldsymbol{y}} \quad & \boldsymbol{b}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}^\top \boldsymbol{y} = \boldsymbol{c} \\ & \boldsymbol{y} \geq \mathbf{0} \end{aligned}$$

This is the dual problem of (1).

General Dual Problems

Primal			Dual		
min	$\boldsymbol{c}^\top \boldsymbol{x}$		max	$\boldsymbol{b}^\top \boldsymbol{y}$	
s.t.	$\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i,$	$i \in M_1,$	s.t.	$y_i \geq 0,$	$i \in M_1$
	$\boldsymbol{a}_i^\top \boldsymbol{x} \leq b_i,$	$i \in M_2,$		$y_i \leq 0,$	$i \in M_2$
	$\boldsymbol{a}_i^\top \boldsymbol{x} = b_i,$	$i \in M_3,$		y_i free,	$i \in M_3$
	$x_j \geq 0,$	$j \in N_1,$		$\boldsymbol{A}_j^\top \boldsymbol{y} \leq c_j,$	$j \in N_1$
	$x_j \leq 0,$	$j \in N_2,$		$\boldsymbol{A}_j^\top \boldsymbol{y} \geq c_j,$	$j \in N_2$
	x_j free,	$j \in N_3,$		$\boldsymbol{A}_j^\top \boldsymbol{y} = c_j,$	$j \in N_3$

- $\boldsymbol{a}_i^\top$ is the ith row of $\boldsymbol{A}$, $\boldsymbol{A}_j$ is the jth column of $\boldsymbol{A}$.
- Each primal constraint corresponds to a dual variable.
- Each primal variable corresponds to a dual constraint.
- Equality constraints always correspond to free variables.

An Example

$$
\begin{aligned}
\text{minimize} \quad & x_1 + 2x_2 \\
\text{subject to} \quad & x_1 + x_2 \geq 5 \\
& x_1 - x_2 \leq 3 \\
& x_1 \geq 0,\ x_2 \in \mathbb{R} \text{ (free)}
\end{aligned}
$$

Constructing the dual problem:

- Associate constraint 1 with dual variable y_1.
- Associate constraint 2 with dual variable y_2.

$$
\begin{aligned}
\text{maximize} \quad & 5y_1 + 3y_2 \\
\text{subject to} \quad & y_1 + y_2 \leq 1 \\
& y_1 - y_2 = 2 \\
& y_1 \geq 0,\ y_2 \leq 0
\end{aligned}
$$

To Memorize: Table & Signs

Primal	minimize	maximize	Dual
constraints	$\geq b_i$	≥ 0	variables
	$\leq b_i$	≤ 0	
	$= b_i$	free	
variables	≥ 0	$\leq c_j$	constraints
	≤ 0	$\geq c_j$	
	free	$= c_j$	

Example

Consider the following problem:

$$
\begin{array}{lrrrl}
\text{minimize} & x_1 & +2x_2 & +3x_3 & \\
\text{subject to} & -x_1 & +3x_2 & & = 5 \\
 & 2x_1 & -x_2 & +3x_3 & \geq 6 \\
 & & & x_3 & \leq 4 \\
 & x_1 \geq 0 & x_2 \leq 0 & x_3 \text{ free} &
\end{array}
$$

The dual is:

$$
\begin{array}{lrrrl}
\text{maximize} & 5y_1 & +6y_2 & +4y_3 & \\
\text{subject to} & -y_1 & +2y_2 & & \leq 1 \\
 & 3y_1 & -y_2 & & \geq 2 \\
 & & +3y_2 & +y_3 & = 3 \\
 & y_1 \text{ free} & y_2 \geq 0 & y_3 \leq 0 &
\end{array}
$$

Today's lecture

- Complexity
- Duality

Next lecture

- Weak and Strong duality

Questions?